Laboratory Exercise 06

Topics

1. Functions design and implementation
    a. Passing Parameters
    b. Use of the statement `return`
2. Stepwise *refinement* to break down complex operations into simpler ones
    a. Design multiple functions to use together
    b. Variables scopes

Discussion

A. What does it mean that the user of a function can consider it as a "black box"?
B. What is the difference, for a function, between returning a value and printing an output value?
C. How can a function be made to be reusable?
D. What is the scope of a variable?

Exercises (in red = suggested ones to complete during the lab)

Part 1 – Single Functions

Goal*: For each of the following exercises, write a Python program that responds to the above requests. Complete <u>at least two</u> exercises during the lab session, and the remaining ones at home.*

[SUGGESTED] 06.1.1 Vowels count. Write the function:

```
def count_vowels(string)
```

that returns the number of vowels present in the string `string`. The vowels are the letters `a`, `e`, `i`, `o`, and `u`, as well as their respective capitalized versions. [P5.6]

[SUGGESTED] 06.1.2 Word count. Write the function:

```
def count_words(string)
```

that returns the number of words in the string `string`. Words are sequences of characters separated by spaces (suppose that there is exactly one space between two consecutive words). For example, `count_words("Mary had a little lamb")` returns `5`.
How could you extend the exercise so that you can properly deal with strings that include multiple spaces between words? [P5.7]

06.1.3 Geometric solids. Write the functions:

```
def sphere_volume(r)
def sphere_surface(r)
def cylinder_volume(r, h)
def cylinder_surface(r, h)
def cone_volume(r, h)
def cone_surface(r, h)
```

that calculate the volume and surface area of a sphere of radius r , of a cylinder with a circular base of radius r and height h , and of a cone with a circular base with radius r and height h. Then write a program that asks the user for the values of r and h , calls the six functions and displays the results as output. [P5.9]

06.1.4 Bank balance. Write a function that calculates the balance of a bank account by crediting interest annually. The function receives as parameters the number of years, the initial balance and the annual interest rate. [P5.22]

Part 2 – Algorithms that make use of functions

Goal*: For each of the following exercises, write a Python program that responds to the above requests. Complete at least two exercises during the lab session, and the remaining ones at home.*

[SUGGESTED] 06.2.1 NGOs. A non-governmental organization needs a program to calculate the share of financial aid to be allocated to each family in need of assistance. The formula is as follows:

I. If the family's annual income is between `$30000` and `$40000` and the family has at least `3` children, the financial aid is equal to `$1000` for each child;
II. If the family's annual income is between `$20000` and `$30000` and the family has at least `2` children, the financial aid is equal to `$1500` for each child;
III. If the family's annual income is less than `$20,000,` the financial aid is `$2,000` for each child.

Write a function that does these calculations. Then write a program that, in a cycle, asks the user to provide the annual income and the number of children of each family applying for the aid, displaying the corresponding value returned by the function. Use `-1` as the sentinel value to finish the data entry. [P5.28]

[SUGGESTED] 06.2.2 Roman numerals. Write a program that converts a Roman numeral, such as `MCMLXXVIII,` to its decimal representation.

Tip: First, write a function that returns the numeric value of each individual letter, then use the following algorithm:

total = 0
s = string corresponding to the Roman numeral
Until s is empty
If s has length 1, or the value of its first character is greater or equal than the value of its second character
Add the value of the first character of s to the total
Remove the first character from s
Otherwise
difference = value of the second character of s - value of the first character of s
Add the difference value to the total
Remove the first two characters from s
[P5.27]

06.2.3 Aerodynamic resistance. The aerodynamic drag force on a car is given by:

$$F_D = \frac{1}{2}\rho v^2 A C_D$$

Where ρ is the density of the air (1,23 kg/m^3), v is the velocity in] m/s , A is the projected area of the car (2,5 m^2) and C_D is the drag coefficient (0,2). The amount of power in watts needed to overcome the resistance force is $P = F_D v$, and the equivalent power in horsepower is $Hp = P/745.7$. Write a program that accepts the speed of the car and calculates the power in watts and horsepower needed to overcome the resulting drag force. [P5.36]

06.2.4 Electric wire. An electrical wire is a cylindrical conductor covered with an insulating material. The resistance of a wire is given by the formula:

$$R = \frac{\rho L}{A} = \frac{4\rho L}{\pi d^2}$$

where ρ is the resistivity of the conductor and L , A and d are the length, cross-sectional area, and diameter of the wire, respectively. The resistivity of copper is $1{,}678 \times 10^{-8}\ \Omega m$. The diameter of the wire, d , is commonly specified by the *American wire gauge (AWG*), as an integer, n . The diameter of an AWG wire n is given by the formula

$$d = 0.127 \times 92^{\frac{36-n}{39}} mm$$

Write a function:

```
def diameter(wire_gauge)
```

that accepts the wire gauge and returns the corresponding diameter. Write another function:

```
def copper_wire_resistance(length, wire_gauge)
```

that accepts the length and gauge of a piece of copper wire and returns its resistance.

The resistivity of aluminum is $2{,}82 \times 10^{-8}\ \Omega m$. Write a third function:

```
def aluminum_wire_resistance(length, wire_gauge)
```

that accepts the length and gauge of a piece of aluminum wire and returns its resistance. Then write a program to test these functions. [P5.35]